

Project 1

Synthetic Cloud and Shadow Generation for Deep Learning-Based Removal from Optical Satellite Images

1. Introduction and Motivation

Optical satellite imagery plays a crucial role in environmental monitoring, agricultural planning, disaster management, urban development, and climate research. However, a major limitation of optical remote sensing is atmospheric occlusion. At any given time, nearly 67% of the Earth's surface is covered by clouds. In addition to cloud cover, cloud shadows further obscure surface structures, reducing the usability of satellite data.

Traditional cloud removal techniques rely heavily on multi-temporal image fusion, where images of the same location taken at different times are combined to reconstruct a cloud-free scene. However, these techniques assume the availability of cloud-free temporal references. In persistently cloudy regions, such data may not exist, making these approaches unreliable.

This thesis addresses this problem by developing a fully synthetic training pipeline combined with a deep learning architecture capable of reconstructing cloud-free satellite images from occluded inputs. Instead of depending on real paired cloudy and cloud-free datasets, which are extremely difficult to obtain, the system generates controlled synthetic clouds and shadows over clean satellite images. These synthetic pairs are then used to train a deep neural network capable of learning contextual reconstruction patterns.

The core innovation of this work lies in the integration of procedural synthetic data generation with a progressive multi-scale attention-based autoencoder architecture.

2. Problem Definition

The problem can be formally defined as follows:

Given an optical satellite image I_{cloudy} that contains cloud and shadow occlusions, reconstruct an image I_{clear} that approximates the true underlying cloud-free surface with high structural and radiometric fidelity.

The primary technical challenges include:

1. Absence of paired training data.
 2. High-resolution reconstruction requirements (up to 1024×1024).
 3. Preservation of structural features such as roads, buildings, and vegetation boundaries.
 4. Avoidance of hallucinated content in heavily occluded regions.
 5. Computational feasibility for large-scale deployment.
-

3. Synthetic Dataset Generation Framework

Since real paired cloudy and cloud-free satellite images are scarce, a synthetic generation framework was developed.

3.1 Clean Image Collection

High-resolution clean optical satellite images were used as the base dataset. These images serve as ground truth targets during supervised training.

3.2 Cloud Mask Generation Using Perlin Noise

Cloud masks were generated using Perlin noise, a procedural noise function commonly used in computer graphics to simulate natural phenomena.

Perlin noise was chosen because:

- It produces smooth, continuous patterns.
- It allows control over scale and frequency.
- It can simulate organic cloud-like structures.

Parameters controlled during generation:

- Coverage ratio (30% to 70%)
- Opacity level
- Cloud scale
- Noise frequency
- Random seed for spatial diversity

For each clean image, 10 synthetic cloudy variants were generated to increase dataset diversity.

3.3 Shadow Projection Modeling

Cloud shadows were generated based on simulated solar geometry.

Parameters used:

- Solar elevation angle
- Solar azimuth angle
- Cloud height approximation

Shadow masks were computed by projecting cloud masks along solar vectors and applying intensity attenuation.

This ensured realistic directional shadows aligned with lighting conditions.

3.4 Blending Process

The final cloudy image was created by blending:

- Original clean image
- Cloud mask
- Shadow mask

Opacity blending ensured smooth transitions rather than hard edges.

3.5 Limitations of Synthetic Approach

Perlin noise cannot simulate:

- Microphysical droplet interactions
- Multi-layer cloud stratification
- Complex atmospheric scattering

Validation indicated 85–90% visual similarity to real cloud patterns. While not physically perfect, the synthetic dataset was sufficiently diverse for robust model training.

4. Deep Learning Architecture – PMAA

The Progressive Multi-scale Attention Autoencoder (PMAA) was implemented as the core reconstruction model.

4.1 Architectural Philosophy

The architecture was designed around three principles:

1. Multi-scale contextual awareness
2. Attention-guided feature weighting
3. Progressive resolution refinement

Instead of performing reconstruction in a single pass, the network gradually refines outputs from low resolution to high resolution.

4.2 Encoder-Decoder Structure

The model follows an encoder-decoder structure.

Encoder:

- Convolutional layers
- Downsampling blocks
- Feature extraction at multiple scales

Decoder:

- Upsampling layers
 - Skip connections
 - Progressive refinement blocks
-

4.3 Multi-scale Attention Module (MAM)

The Multi-scale Attention Module computes both spatial and channel attention.

Spatial Attention:

$$A_{\text{spatial}}(x) = \text{Sigmoid}(\text{Conv}(\text{ReLU}(\text{Conv}(x))))$$

Channel Attention:

$$A_{\text{channel}}(x) = \text{Sigmoid}(\text{FC}(\text{ReLU}(\text{FC}(\text{GAP}(x)))))$$

Where:

- GAP = Global Average Pooling
- FC = Fully Connected Layer

Final attention-weighted output:

Output = $x \times A_{\text{spatial}} \times A_{\text{channel}}$

This mechanism allows the model to:

- Suppress cloud-affected regions
 - Emphasize structurally reliable areas
 - Capture long-range contextual dependencies
-

4.4 Local Interaction Module (LIM)

The LIM enhances fine-grained texture reconstruction.

It combines:

- Local convolutional features
- Skip connections from earlier encoder layers
- Residual blocks

This preserves:

- Building edges
 - Road networks
 - Vegetation texture
-

4.5 Progressive Refinement Strategy

Training was performed in stages:

Stage 1 → 256×256 resolution

Stage 2 → 512×512 resolution

Stage 3 → 1024×1024 resolution

Each stage initializes weights from the previous stage.

This approach:

- Stabilizes training
 - Reduces memory overhead
 - Allows gradual complexity increase
 - Improves convergence
-

5. Training Strategy

5.1 Loss Functions

The generator was trained using a composite loss:

$$L_{\text{total}} = L_{\text{adv}} + \lambda_1 L_1 + \lambda_2 L_{\text{perceptual}}$$

Where:

L_{adv} = Adversarial loss (binary cross-entropy)

L_1 = Pixel-level reconstruction loss

$L_{\text{perceptual}}$ = VGG-based feature-space loss

Hyperparameters:

$$\lambda_1 = 10$$

$$\lambda_2 = 0.1$$

Optimizer:

Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)

Learning rate:

$$2 \times 10^{-4}$$

Batch size:

8

Early stopping based on validation SSIM.

Total training time:

~40 hours on NVIDIA A100 GPU.

6. Evaluation Methodology

All models were trained under identical conditions for fair comparison.

Evaluation metrics:

- PSNR (Peak Signal-to-Noise Ratio)
- SSIM (Structural Similarity Index)
- MSE (Mean Squared Error)
- RMSE

- MAE
- Inference Time

Statistical rigor:

- 5 independent runs
 - 95% confidence intervals
 - Paired t-tests ($p < 0.001$)
-

7. Results

PMAA significantly outperformed CTGAN and STGAN.

PSNR:

35.08 dB (PMAA) vs 22.30 dB (CTGAN) vs 11.88 dB (STGAN)

SSIM:

0.9673 (PMAA) vs 0.879 (CTGAN) vs 0.529 (STGAN)

Ablation results:

Removing attention:

−4.8 dB PSNR drop

Removing progressive refinement:

−6.2 dB PSNR drop

This confirms architectural contributions are statistically significant.

8. Computational Analysis

For 1024×1024 images:

- 12.3 GFLOPs per inference
- 0.0341 seconds per image
- 29 images/sec in batch mode
- 8.2GB peak GPU memory

Quantization reduced memory by 60% with negligible PSNR loss (<0.5 dB).

This demonstrates deployment feasibility.

9. Risk Assessment

Potential risks:

- Hallucinated reconstructions in dense cloud regions
- Misinterpretation in disaster scenarios
- Overreliance without uncertainty estimation

Mitigation recommendations:

- Confidence scoring
 - Human verification
 - Multi-model ensemble validation
 - Documentation of uncertainty bounds
-

10. Skills Demonstrated

- Synthetic data engineering
 - Computer vision
 - Attention mechanism implementation
 - Deep learning architecture design
 - Statistical validation
 - GPU optimization
 - Model compression
 - Ethical AI analysis
 - Research documentation
-

This is now the level of depth suitable for a standalone Google Docs file.

Next project:

Say:

“Continue with Adaptive Resilience”

And I will write that one in the same in-depth format.